

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

- Q1.** Automated Document Processing System A document processor handles one page at a time using the recurrence $T(n) = T(n-1) + 1$. Recall the meaning of a recurrence relation, explain how it expands step-by-step, and solve it using substitution method to determine the final time complexity.
- Q2.** Also determine the space complexity of the recursive process.

Multimedia Streaming Platform A streaming platform divides media data into two equal parts and processes them recursively, with extra linear work at each stage: $T(n) = 2T(n/2) + n$. State the Master Theorem, explain the significance of a, b, and f(n), and apply it to determine the time complexity. Also analyze the space complexity of this recursive streaming process.

Code of Conduct

I, K. Jayadevkata Sai (Name / Reg No) 192525182 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Jayadevkata Sai

Questions 4 and 5

Question 4: Automated Document Processing System

Question

A document processor handles one page at a time using the recurrence $T(n) = T(n-1) + 1$. Recall the meaning of a recurrence relation, explain how it expands step-by-step, and solve it using the substitution method to determine the final time complexity. Also determine the space complexity of the recursive process.

Meaning of a Recurrence Relation

A recurrence relation defines the running time of an algorithm in terms of smaller input sizes. Here, $T(n)$ is the time to process n pages, $T(n-1)$ is the time to process $n-1$ pages, and $+1$ is the time to process one additional page.

Expansion

$$T(n) = T(n-1) + 1$$

$$= T(n-2) + 2$$

$$= T(n-3) + 3$$

...

$$= T(1) + (n-1)$$

$$\text{Since } T(1) = 1,$$

$$T(n) = n$$

Substitution Method

$$T(n) = T(1) + (n-1) = 1 + (n-1) = n$$

Therefore, Time Complexity = $O(n)$.

Time Complexity

Best = $O(n)$, Average = $O(n)$, Worst = $O(n)$.

Space Complexity

Recursive call stack = $O(n)$.

Conclusion

Final Time Complexity = $O(n)$, Space Complexity = $O(n)$.

Question 5: Multimedia Streaming Platform

Question

A streaming platform divides media data into two equal parts and processes them recursively, with extra linear work at each stage: $T(n) = 2T(n/2) + n$. State the Master Theorem, explain the significance of a , b , and $f(n)$, and apply it to determine the time complexity. Also analyze the space complexity of this recursive streaming process.

Master Theorem

$T(n) = aT(n/b) + f(n)$, where a is the number of subproblems, b is the division factor, and $f(n)$ is the extra work.

Values

$a=2$, $b=2$, $f(n)=n$, and $n^{\log_b a}=n$.

Application

Since $f(n) = \Theta(n) = \Theta(n^{\log_b a})$, Case 2 of the Master Theorem applies.
Therefore, $T(n) = \Theta(n \log n)$.

Time Complexity

Best = $O(n \log n)$, Average = $O(n \log n)$, Worst = $O(n \log n)$.

Space Complexity

Recursion depth = $O(\log n)$.

Conclusion

Final Time Complexity = $\Theta(n \log n)$, Space Complexity = $O(\log n)$.